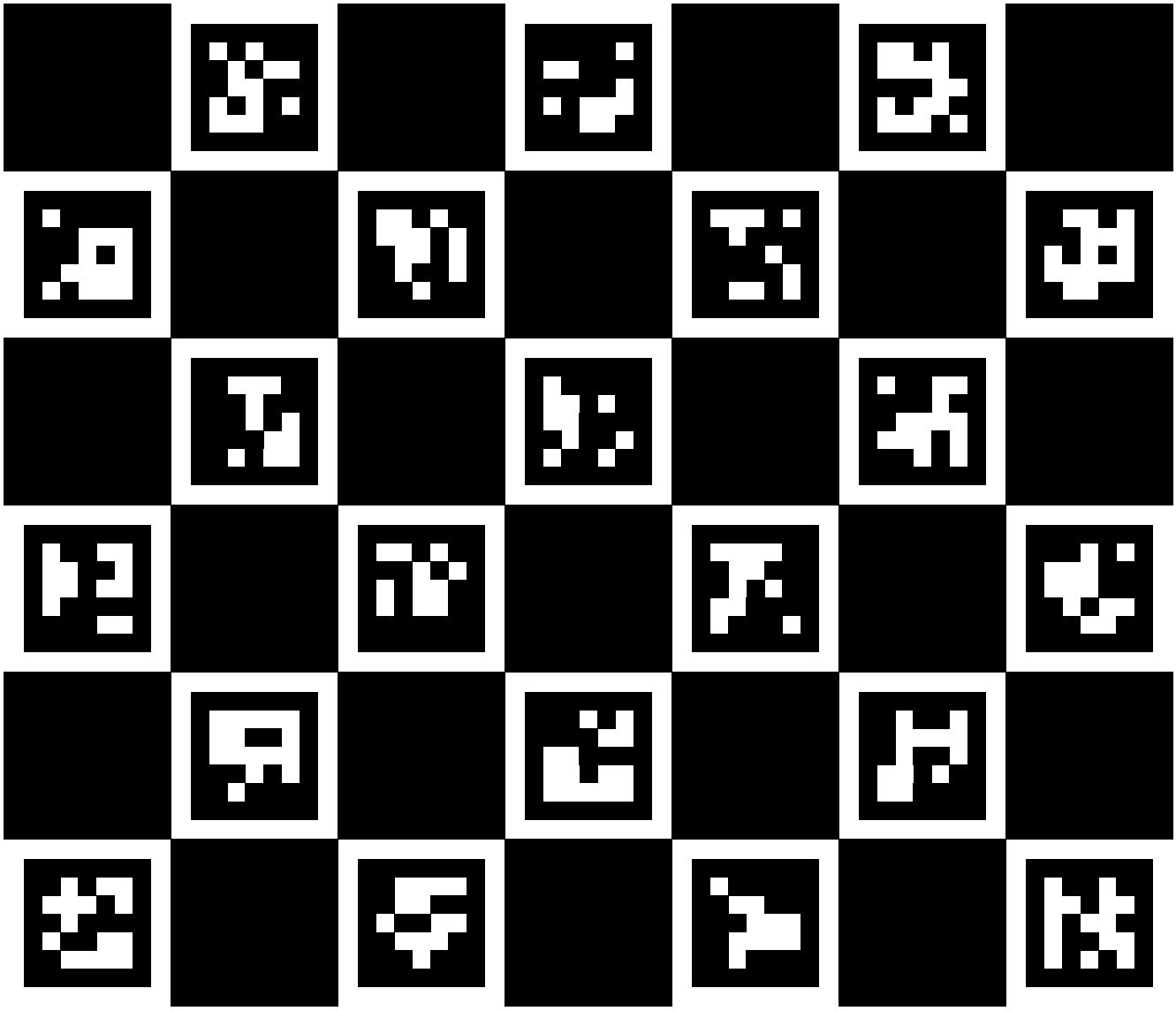
ChArUco metric reference board

SQUARE 25.00 mm

marker 19.00 mm | grid 7 x 6 | DICT_5X5_1000 | board 175 x 150 mm | 21 markers, 30 interior corners

PRINT AT 100% / "ACTUAL SIZE". Turn OFF "Fit to page", "Shrink to fit" and any auto-rotate. Matte paper, not glossy.

These dimensions are NOMINAL - what was drawn, not what your printer produced. Measure the bars below before this sheet is used for anything. *A4-7x6-25mm-DICT_5X5_1000-7D0B57C9*



125.00 mm (X)

measure the full length of this black bar, end to end

AFTER PRINTING - do this before you shoot anything

1. Digital calipers across the full length of the X bar above. Write it down to 0.01 mm.
2. Same for the Y bar down the right edge. Two axes, because a printer can scale them differently.
3. Put both readings into config/charuco.json:

   `"measured_span_x_mm": <your X reading>, "measured_span_y_mm": <your Y reading>`
4. Fill in measured_on / measured_by / paper too, so a future reprint is not confused with this sheet.

Why: a printer that quietly scales to fit puts that error straight into every dimension the pipeline reports.

A reading more than ~0.5% off nominal, or X and Y disagreeing by more than 0.5%, means reprint - do not average it out.

Toner spreads roughly 0.1 mm on a bar edge. That is far below the print-scaling error this is correcting; ignore it.

Keep the sheet flat - tape it down or mount it on board. A curl is a scale error the calipers will not catch.

Full instructions: CALIBRATION_NOTE.md

75.00 mm (Y)